



SIMATS Online Programming Lab

C PROGRAMMING



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CO5 - AT1 - LEVEL-3

Level Exam

C

02 Sep 2026, 02:45 PM

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Result Available

Your score: **85 / 100**

QUESTIONS

LEVEL 3 — HARD

Q1

function

Q2

functionsandpointers

2 / 2 saved

Level 3 — Hard **sort (H)** — function

Write a function `void mergeSortedArrays(int arr1[], int n1, int arr2[], int n2, int merged[])` that merges two sorted arrays `arr1` and `arr2` into a single sorted array `merged`.

Sample Input:

Enter the size of first array: 3

Enter the elements of first array: 1 3 5

Enter the size of second array: 3

Enter the elements of second array: 2 4 6

Sample Output:

Merged sorted array: 1 2 3 4 5 6

Your Code

```
1 #include <stdio.h>
2
3 void mergeSortedArrays(int arr1[], int n1
4     int i = 0, j = 0, k = 0;
5
6     while (i < n1 && j < n2) {
```

```
7         if (arr1[i] <= arr2[j]) {
8             merged[k++] = arr1[i++];
9         } else {
10            merged[k++] = arr2[j++];
11        }
12    }
13
14    while (i < n1) {
15        merged[k++] = arr1[i++];
16    }
17
18    while (j < n2) {
19        merged[k++] = arr2[j++];
20    }
21 }
22
23 int main() {
24     int n1, n2;
25
26     printf(" Enter the size of first array\n");
27     scanf("%d", &n1);
28
29     int arr1[n1];
30     printf("\n Enter the elements of first array\n");
31     for (int i = 0; i < n1; i++) {
32         scanf("%d", &arr1[i]);
33     }
34
35     printf("\n Enter the size of second array\n");
36     scanf("%d", &n2);
37
38     int arr2[n2];
39     printf("\n Enter the elements of second array\n");
40     for (int i = 0; i < n2; i++) {
41         scanf("%d", &arr2[i]);
42     }
43
44     int merged[n1 + n2];
45     mergeSortedArrays(arr1, n1, arr2, n2,
46
```

```
47     printf("\n Merged sorted array: ");
48     for (int i = 0; i < n1 + n2; i++) {
49         printf("%d ", merged[i]);
50     }
51     printf("\n");
52
53     return 0;
54 }
```

Program Input (stdin)

Enter input if required...

Output

Enter the size of first array:
Enter the elements of first array:
Enter the size of second array:
Enter the elements of second array:
Merged sorted array: 1 2 3 4 5 6

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